Ultrasonic Cutting — Critical Actions

U.S. Solid 28 kHz / 500 W handheld · Fybron, FyberCom, Fyberite

An ultrasonic blade is part of the resonant stack, not a generic consumable. **A 24 kHz blade will not run on the 28 kHz machine.** Buy for the specific unit.

1 · Order now — approx. $400 total

Qty	Item	Where	Each	Purpose
3	U.S. Solid QG-100 replacement blade, titanium alloy	ussolid.com search *QG-100 blade*	**$126**	one factory control, one to relieve, one to coat — see §3
1	DLC coating service on one blade	tgwint.com x-keenblades.com	quote	~80 HRC, low friction — cuts flank heat and survives Fyberite's carbon

Confirm QG-100 is your machine before ordering — U.S. Solid's cutter listing never states the model — and ask for a blade drawing. They publish no dimensions, thickness or edge angle, so there is currently nothing to specify against.

2 · Do — to stop the lyocell scorching

Recondition the stock before cutting. Never cut straight off the press.	Moisture absorbs ~1.9x the energy before char. Pressing at 160–180 °C leaves it hot and bone dry — the worst case. Free to fix, and the biggest single lever.
Go faster on Fybron and FyberCom. Slower (1 m/min) on Fyberite.	Cellulose damage is time-at-temperature. Slowing down when you see scorch makes it worse. Fyberite is the opposite case — no burn risk, only a mechanical limit.
Keep the blade sharp. Relieve and polish the flanks.	Most heat is flank rubbing, not tip cutting. A dull blade rubs instead of cutting — pure heat, no progress.
Judge by tear strength against a die-cut control, not by edge colour.	Strength loss precedes visible browning. A "looks fine" edge can already be degraded.

3 · What to do with the three blades

Blade	Prep	What it tells you
A — control	none, as delivered	baseline edge quality and scorch threshold on all three materials
B — relieved	hand back-bevel the flanks with fine stones, then polish, so only the edge contacts the kerf	the highest-leverage change available to the tool you already own; isolates flank friction as the heat source
C — DLC	send out for diamond-like carbon coating	friction and wear together — the only one worth running on carbon-loaded Fyberite for a tool-life number

Coatings add microns and are safe. Grinding removes mass and shifts resonance — relieve conservatively and confirm the generator still locks.

4 · Also worth $30

A solid tungsten-carbide rotary knife (Sollex Z50, Ø25 mm) cuts cellulose **cold**, with almost no heat. At ~$30 it is the cheapest possible test of whether Fybron needs to be ultrasonic at all — run it alongside the blade trials. sollex.com — *Knives to Cut Carbon, Aramid & Glass Composites*

References

Source	Reference	Used here for
Kim, H.G., Hong, T.H., Kim, D., Kim, S.H. (2024)	An experimental study of ultrasonic-knife cutting for a woven carbon fiber preform by an industrial robot. *Manufacturing Letters* 41:581–587 (NAMRC 52).	traverse speed 1 m/min; 45° attack angle; 24° cemented carbide; 1,200 W / 24 kHz / 60 µm
Kara, S. (2024)	Ultrasonic weldability of thick and heavier nonwoven fabrics. *J. Appl. Polym. Sci.* 141(33):e55840. doi:10.1002/app.55840.	60–65% thermoplastic threshold; inefficient energy transfer above 0.69 mm; strain loss 66% –> 10–12%; 108.9–227.9 gsm ceiling
Lee, C.H., Seo, J.S., Park, D.S. (2014)	Design and Tuning of the Ultrasonic Wide-blade Horn for Joining and Cutting Non-woven Fabrics. *Adv. Mater. Res.* 941–944:1932–1936. doi:10.4028/www.scientific.net/AMR.941-944.1932.	amplitude uniformity 76% across a 180 mm blade; tool-steel horn; slot geometry
Dukane AN503 (rev. 02, 2016)	Cutting Food Products or Materials with Ultrasonic Blade Horns.	amplitude 70–100%; ramp-up 0.150 / 0.100 / 0.050 s at 20 / 30 / 40 kHz
Dukane AN512	Stack Scan.	locating free-running frequency — the fault mode behind the tuning warning above
US 4,560,427 (1985)	Ultrasonic seal and cut method and apparatus.	"typically 50 percent or more" thermoplastic fibre to cut and seal; 16–100 kHz
US 4,542,771 (1985)	Adjustable anvil for ultrasonic material cutting and sealing apparatus.	anvil sealing surface 3–20° included, preferably 5–15°; reinforcing ~30°
US 5,265,508 (1993)	Ultrasonic cutting system for stock material.	shoe-to-anvil clearance ~0.003 in
Internal	*Material Samples_Formulations & Testing.xlsx*, July 2026 runs.	gsm, thickness, density, tensile, tear; press schedules 160 °C Fybron / 180 °C Fyberite

Full text of every source above is held in the RAGFlow dataset "Ultrasonic Cutting" (18 documents, 177 chunks). Vendor pages: ussolid.com · tgwint.com · x-keenblades.com · sollex.com · sonotec.com/en/product/tool_blade/ · artcotools.com